Sultanate of Oman
Ministry of Health
The Royal Hospital
Department of Child Health

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Title: Approach to a Child with Global Developmental Delay

Statement:

The approach to a child with global developmental delay (GDD)protocol is used to guide clinicians on the investigations and referral criteria for all children with GDD within the pediatric population.

Definition:

Global developmental delay is defined as significant delay in two or more developmental domains; gross and fine motor; speech and language;cognitive, personal and social development; or activities of daily living.

Specific developmental delay refers to delay in single domain (suchas gross motor delay or delay in speech and language)

"Significant" may be defined as performance 2 or more standard deviations below the mean on developmental screening or assessment tests.

Global developmental delay refers to children <5yrs of age and Mental retardation (USA) or Learning disability/Intellectual disability (UK) is used for children >5yrs of age.

Classification:

Mild delay if functional age is <33% below chronological age

Moderate delay if functional age is 34–66% of chronological age

Severe delay if functional age is >66% below chronological age

There are several disabilities in the classification of developmental delay:

· Gross motor delay

· Developmental language disorders

· Global developmental delay

· Visual sensory impairment

· Hearing sensory impairment

· Sensory Processing disorders

· Learning disabilities

· Autism Spectrum disorders

Prevalence:

· The precise prevalence of global developmental delay is unknown.

· Estimates of 1% to 3% of children younger than 5 years of age in the general population.

Presentation:

· Through routine surveillance or developmental screening in some countries.

· Children with identified risk factors (such as prematurity) may be identified early through follow up.

· Parental concerns on child's developmental achievement.

· Abnormal patterns of development may be detected by professionals in a nursery or day care setting,raising their concerns to the family.

· Concerns may be detected incidentally on an acute presentation to the hospital, if questions are asked about development.

Developmental Screening:

Screening is a process to identify children at increased risk of having developmental difficulties that uses relatively brief and simple techniques, according to well recognized criteria.

It is recommended by the AAP that every childshould have developmental screening between the ages of 18-24months.

The screening program can be done by the primary health centers or by any health care worker who encounters an opportunity with a child.

Benefits:

- Early diagnosis and intervention.

- Early diagnosis of conditions with a genetic basis, such as fragile X syndrome, facilitates genetic counseling for families.

- Provides parents with reliable information before a developmental problem becomes obvious and gives them more time to adjust to the child's difficulty and make appropriate management plans for their family.

- Parents are reassured and relieved of anxiety if assessment shows that the child is within the normal range.

- Early assessments can be compared with later ones, allowing the practitioner to follow a child's individual developmental trajectory

- Provides an opportunity to encourage good parenting and developmental stimulation.

Suggested opportunistic screening questions for any age child:

- Do you have any concerns about the way your child is behaving, learning, or developing?

- Do you have any concerns about the way he or she moves or uses his or her arms or legs?

- Do you have any concerns about how your child talks and understands what you say?

- Does your child enjoy playing with toys? Describe what he or she does while playing

- Does your child get along with others?

- Do you have anyconcerns about how your child is learning to do things for him or herself?

One of the commonly used screening tool for developmental delay is Ages and Stages questionnaire, others include MCHAT R and MCHAT F/U which are used to screen children for autism spectrum disorder.

History:

Family history • Three generations, maternal and paternal • Consanguinity • Previous pregnancy outcomes: miscarriages, stillbirths, neonatal or childhood deaths, infertility • Family history of birth defects, childhood deaths, mental retardation, speech delay, learning disabilities, autism and known genetic conditions – note level of education in parents and siblings • Ethnic background	**Pre & Perinatal history** • Potential teratogens including alcohol, medications, vitamins, maternal infection (rubella, cytomegalovirus, toxoplasmosis, varicella), maternal diabetes, hyperthermia, maternal phenylketonuria. • Fetal movements • Prenatal tests (eg, amniocentesis, ultrasound) • Gestation, mode of delivery, Apgar scores, resuscitation • Birth weight, length, head circumference • Feeding, muscle tone, other problems
Postnatal history • Milestones, school performance • Evidence of regression • Unusual behavior, personality • Coordination, seizures, unusual movements, increased or decreased tone • Previous illnesses • Vision, hearing	**Social History:** • Evidence of neglect or abuse which may have a negative influence on development. • Primary languages.

pediatrician.

- Loss of developmental skills at any age
- No speech by 18 months, especially if the child does not try to communicate by other means such as gestures (simultaneous referral for urgent hearing test)
- Asymmetry of movements or other features suggestive of cerebral palsy, such as increased muscle tone
- Persistent toe walking
- Complex disabilities
- An assessing clinician who is uncertain about any aspect of assessment but thinks that development may be disordered
- Sit unsupported by 12 months
- Walk by 18 months (boys) or 2 years (girls) (check creatine kinase urgently)
- Run by 2.5 years
- Hold object placed in hand by 5 months (corrected for gestation)
- Reach for objects by 6 months (corrected for gestation)
- Point at objects to share interest with others by 2 years

Clinical Examination:

Growth Parameters:

- Microcephaly: eg in Rett's Disorder
- Macrocephaly: eg in hydrocephalus
- Short stature: Turner syndrome, Williams syndrome
- Obesity: Prader-Willi syndrome, Beckwith-Wiedemann syndrome

Genitourinary

- Macroorchidism: Fragile X syndrome
- Hypogonadism: Prader-Willi syndrome

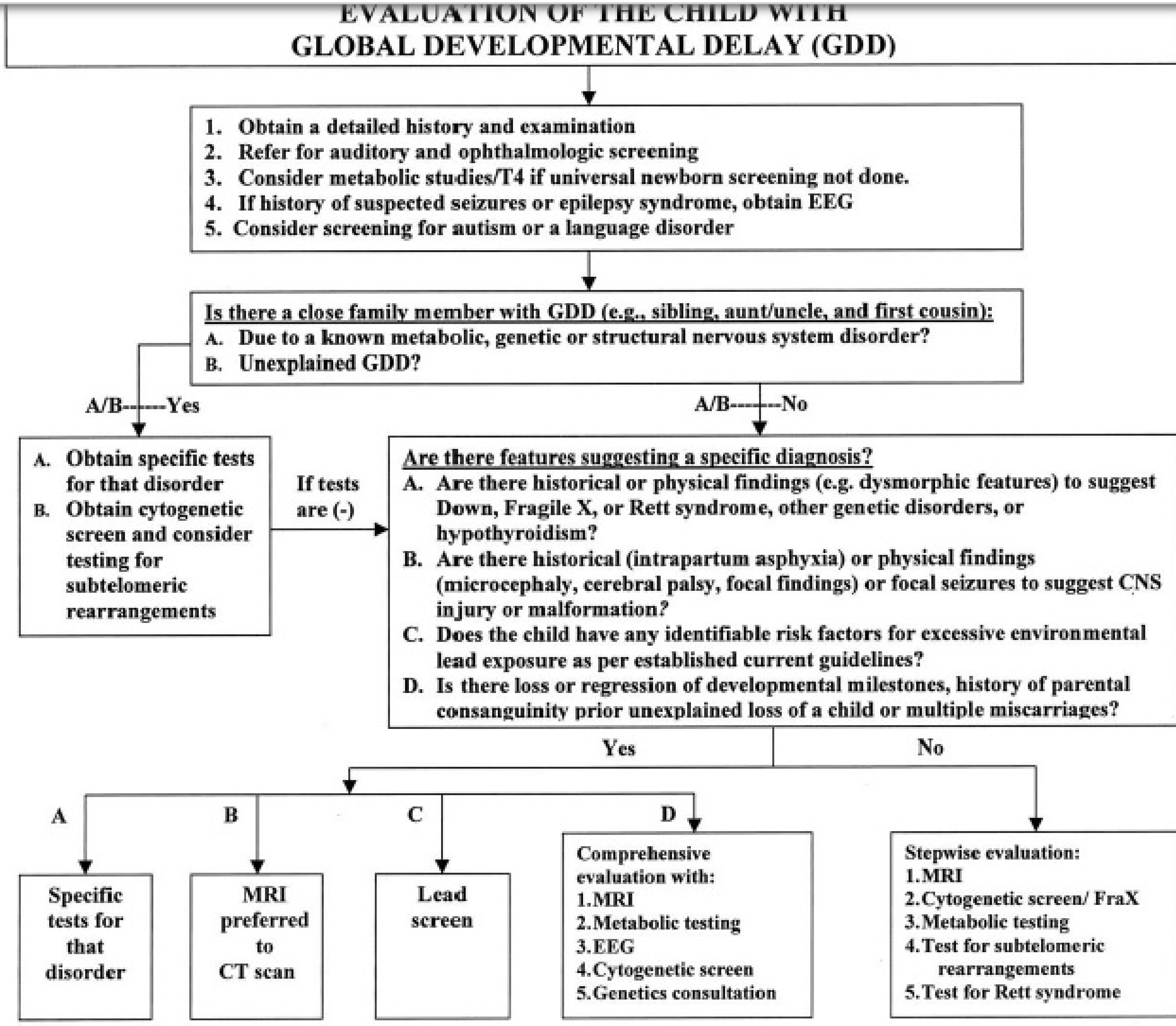
Extremities

- Small hands: Prader-Willi syndrome
- Clinodactyly: trisomies including Down syndrome
- Transverse palmer crease: Down syndrome

Head and Neck

- Flat occiput: Down syndrome, Zellweger syndrome
- Prominent occiput: trisomy 18
- Craniosynostosis: Crouzon syndrome, Pfeiffer syndrome
- Midface hypoplasia: Fetal Alcohol Syndrome (FAS), Down syndrome
- Prominent nose and chin: Fragile X syndrome
- Round facies: Prader-Willi syndrome
- Triangular facies: Turner syndrome
- Hypertelorism: Fetal hydantoin syndrome
- Hypotelorism: maternal PKU effect
- Brushfield spots: Down syndrome
- Prominent eyes: Beckwith-Wiedemann syndrome
- Lisch nodules: neurofibromatosis
- Large pinna: Fragile X syndrome
- Malformed pinna: Treacher Collins syndrome, CHARGE association
- Broad nasal bridge: Fragile X syndrome
- Low nasal bridge: Down syndrome
- Long philtrum: FAS
- Cleft lip and palate: may either be isolated or part of a syndrome
- Micrognathia: Robin sequence
- Macroglossia: Beckwith-Wiedemann syndrome
- Abnormal hair whorls: Down syndrome
- Webbed neck: Turner syndrome

• Nail hypoplasia or dysplasia: FAS • Facial port wine hemangioma: Sturge-Weber syndrome • Café au lait spots: Neurofibromatosis • Ashleaf spots: Tuberous Sclerosis	• Cranial nerves • Specific vision tests: red reflex, normal fundi, response to visual stimuli, field of vision • Specific auditory tests: response to auditory stimuli • Receptive or expressive language delay • Abnormal speech (eg. articulation) • Persistently present Babinski response (older than 2 years of age) • Hyper- or Hypotonia • Sensory • Motor strength • Gait • Deep tendon reflexes • Primitive reflexes – Moro, Gallant • Postural reflexes – propping response



Intervention:

- Early identification and early intervention is crucial and has a good outcome for children with developmental impairments.
- Identified patients needs to be referred to the appropriate services/therapies such as physiotherapy, speech and language therapy, occupational therapy, psychologist, early learning interventions etc.
- Patients should be assessed on all areas to identify what kind of disabilities they have and what kind of support to be offered.
- Other treatments modalities can be tailored according to the diagnosis reached

Title of book/ journal/ articles/ Website	Author	Year of publication	Page
Arch Dis Child 2006;91:701–705. doi: 10.1136/adc.2005.078147 http://www.ncbi.nlm.nih.gov/pmc/articles/PMC2083045/pdf/701.pdf		2005	
Evaluation of the child with GDD, AAN guideline summary for clinician. http://tools.aan.com/professionals/practice/guidelines/guideline_summaries/Global_Devlopmental_Delay_Clinicians.pdf		2011	
Developmental assessment of children, Clinical Review, *BMJ* 2013;346:e8687 doi: 10.1136/bmj.e8687 (Published 15 January 2013). http://www.bmj.com/content/bmj/346/bmj.e8687.full.pdf		2013	
Developmental delay: Causes and investigation by Angharad V Walters, published in Paediatric Neurology Journal, ACNR, volume 10 number 2, May/June 2010. http://www.acnr.co.uk/may_june_2010/ACNRMJ10_Developmental.pdf		2010	
The child with developmental delay: An approach to etiology, published in Paediatr Child Health Vol 8 No 1 January 2003. http://www.ncbi.nlm.nih.gov/pmc/articles/PMC2791071/pdf/pch08016.pdf		2003	
Basics to the approach of developmental delay http://learn.pediatrics.ubc.ca/body-systems/nervous-/syste/basics-to-the-approach-of-developmental-delay		2011	

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